PhaseAtlas
Submission kit & presentation scripts

Niu Te | Imperial College London
18 October 2026 | Pre-event project snapshot

Submission requirements

Item	2026 requirement
Deadline	22 October 19:00 CDT = 23 October 00:00 UTC
Platform	Moment: scientificintelligence.org/onboarding?campaign=osi-hackathon-2026
Team	Every member signs in and joins the same project team; one member submits.
Video	Up to four minutes; retain natural speech speed.
Public post	Video must be viewable without logging in.
Write-up	One page, ideally with a figure.
Code	Provide the repository link.
Platforms	Select the platforms actually used; sponsor awards require use of that platform.

Prepared files

PhaseAtlas_One_Page.pdf is the current one-page summary with seven plots, including 3D space-time visualization. The narrated film is 3 minutes 38 seconds. The article and scientific case arrays provide supplementary methods and results.

Submission record

Use the current Moment entry form, review the links in a signed-out browser and retain the final confirmation. This package contains source and supporting material; the participant completes the platform submission.

Prior work

Describe the project history and event-period changes according to the organizer requirements and the actual development record.

2. Copy-ready project fields

Field	Suggested entry
Project / team name	PhaseAtlas
Author / team member	Niu Te
Affiliation	Imperial College London
Area	Scientific software; mathematics; computational physics
Title	PhaseAtlas: From PDE Animations to Reproducible Numerical Evidence
Source URL	https://github.com/NIU-55383/PhaseAtlas
Existing host	https://phaseatlas.onrender.com/ - verify the final release before submission
Development / runtime AI	AI-assisted development; no external LLM call during numerical simulation

Short description

PhaseAtlas is a browser workbench that turns editable PDE models into animated, inspectable experiments. It combines controlled refinement, boundary-balance checks, travelling-wave diagnostics and parameter maps with reusable experiment records. Robin boundary cooling provides an application example; a fourth-order front connects the platform to nonlinear-wave research. The focus is reproducible numerical evidence rather than a claim to solve every PDE.

Contribution / differentiation

The project combines three research tasks in one workflow: checking numerical resolution and provenance, separating wave motion from deformation, and mapping a user-defined response while preserving every run. These individual methods are established; the contribution is their independent implementation and integration. Ninety mathematical and numerical tests pass in the current snapshot. No quantified superiority over another solver or proven user productivity gain is claimed.

Prior-work disclosure

The original Fourth-order mathematics and the 18 October 2026 software baseline were developed before the event. The package records that baseline. Describe only changes actually completed on October 21-22 as event-period work, with dated commits or an equivalent change record. The current article and summary do not claim work occurred during a future event.

Fields requiring a final human entry

The confirmed 2026 submission form; your contact email if requested; the public video/post URL after publication; a verified URL for the newly deployed release; and a factual event-period contribution statement. No form was submitted and no message was sent on your behalf.

3. Four-minute presentation script

Target: about 3 minutes 55 seconds. Rehearse and measure the final recording; edit pauses to stay below the announced limit. Pre-run studies so solver waiting time does not consume the presentation. Use the local reviewed release until the same version is deployed.

0:00-0:35 | The problem

Screen: Homepage: show the real cooling animation.

A PDE animation can look convincing even when its boundary conditions or numerical resolution are wrong. I built PhaseAtlas to connect the picture to the evidence behind it. It is a browser workbench for researchers who want to edit a model, inspect its dynamics and share a calculation that another person can check.

0:35-1:15 | A practical question

Screen: Cooling case: open its model in the workbench, run and play.

Start with a hot rod exchanging heat with its surroundings. The Robin boundary condition represents that exchange. I can change the diffusion coefficient or the exchange coefficient, run the equation and inspect the temperature through time. The model, initial data and boundary conditions remain visible. This is a dimensionless model study; I am not claiming it predicts a particular material without calibration.

1:15-1:55 | Check the calculation

Screen: Robin benchmark: run the three-space-level study.

To test the boundary implementation, I use a heat-equation mode with a known exact solution. I hold the time step fixed and refine the grid from sixteen to thirty-two to sixty-four cells. The measured errors decrease, and the observed order is approximately two. A separate balance audit compares the thermal-content change with the input through the walls. Neither a smooth animation nor a small algebraic residual is used as an accuracy certificate.

1:55-2:35 | Compare a family of experiments

Screen: Show the cooling parameter map; select one cell and replay.

Now I turn one run into a parameter question: how much heat remains after a fixed time? Each map cell stores a real numerical trajectory. I can choose a mean, a local probe or a threshold event. Failed runs stay visible, and an event that was not observed is left missing. This is a finite-time parameter scan, not a bifurcation diagram.

2:35-3:15 | The original research connection

Screen: Front laboratory: play disturbance map and point to norm curves.

The second case comes from my earlier work on a fourth-order travelling front. PhaseAtlas retains that mathematical structure while letting me distinguish motion from deformation. Global, local and weighted norms can behave differently. The general Studio also lets a researcher specify two linear boundary conditions at each end for supported fourth-order equations. These calculations support research questions; they do not prove nonlinear stability.

3:15-3:55 | Reuse and contribution

Screen: Evidence export, article and reproducibility download.

The contribution is a connected workflow: editable equations, visual experiments, numerical checks and a portable research record. The present suite passes ninety mathematical and numerical tests, with an independent Python reference for selected problems. Computation runs on the user's device. I used AI assistance in development, while the solver itself is deterministic and needs no language-model API. The package identifies its pre-event baseline so later hackathon contributions can be recorded honestly. I am Niu Te from Imperial College London. This is PhaseAtlas: from PDE animations to reproducible numerical evidence.

4. Two-minute alternative

Use if the organizer confirms the shorter legacy limit. Target: about 1 minute 55 seconds; time the actual narration.

0:00-0:25

Screen: Show homepage cooling animation.

A persuasive PDE animation does not tell us whether a computation is resolved. PhaseAtlas connects an editable model to visual experiments, numerical checks and an exportable research record. I am Niu Te from Imperial College London.

0:25-0:55

Screen: Show Robin model and refinement chart.

A hot rod exchanging heat with its surroundings gives a practical starting point. Robin boundary data represent wall exchange. A separate exact heat-equation mode lets us test the numerical implementation: on sixteen, thirty-two and sixty-four cells, errors decrease with observed order approximately two. A balance audit exposes the relationship between heat leaving the walls and thermal-content change.

0:55-1:25

Screen: Show parameter map and original wave animation.

Every cell in a parameter map retains its trajectory and metric definition. A researcher can compare cooling responses and revisit a failed run. The original fourth-order front from my earlier research adds wave-motion and deformation diagnostics, with configurable extra boundary conditions in the general Studio. The numerical evidence is not presented as a stability theorem.

1:25-1:55

Screen: Show JSON export and article.

Another researcher receives equations, boundaries, arrays and method information, plus selected independent Python checks. The current suite passes ninety tests. The implementation is independent and AI-assisted; simulation runs locally without a runtime language-model API. The dated package separates prior work from contributions made during the event. PhaseAtlas helps turn an interesting picture into an experiment that someone else can inspect.

5. Final submission checks

1. Confirm the current form and prior-work policy from the organizer. Register every actual team member.
2. Record the actual event-period changes separately from this baseline.
3. Deploy the reviewed source and open the demo, PDF and JSON links in a signed-out browser.
4. Record and time the presentation. This package supplies scripts, not a finished narrated video.
5. Publish the intended social post/video only after reviewing its content and URLs.
6. Complete the official form if required and retain its confirmation before the deadline.

Detailed social copy is provided as social-post.txt. This kit gives suggested fields; the current 2026 form schema has not been independently confirmed. Team size: the official FAQ recommends 3-10 but imposes no size limit; do not describe three as a required minimum.